

Elements of Design

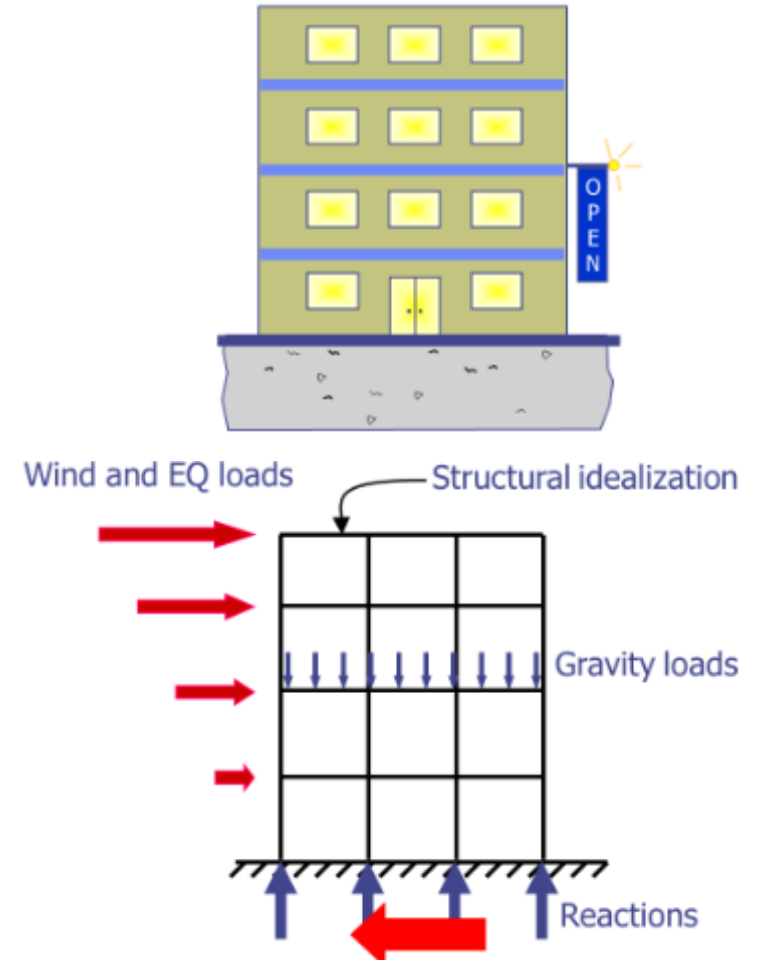
Hector Garcia Matamoros

University of California, Berkeley

August 2026

Contents

1. Introduction
2. Design loads
3. Structural Design

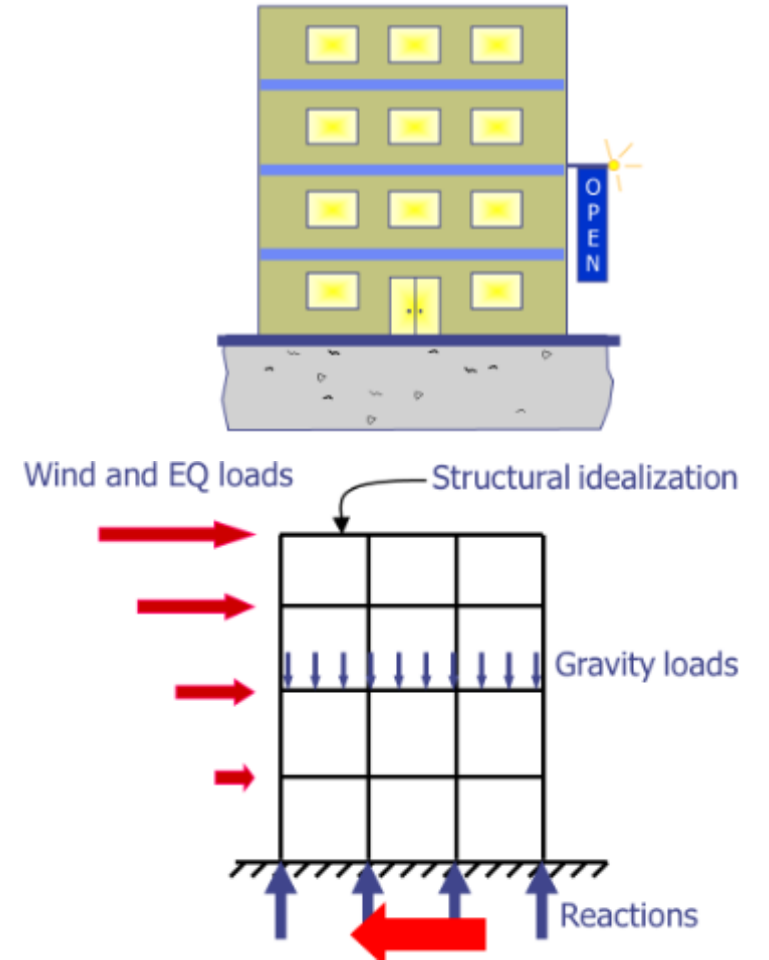


Contents

1. Introduction

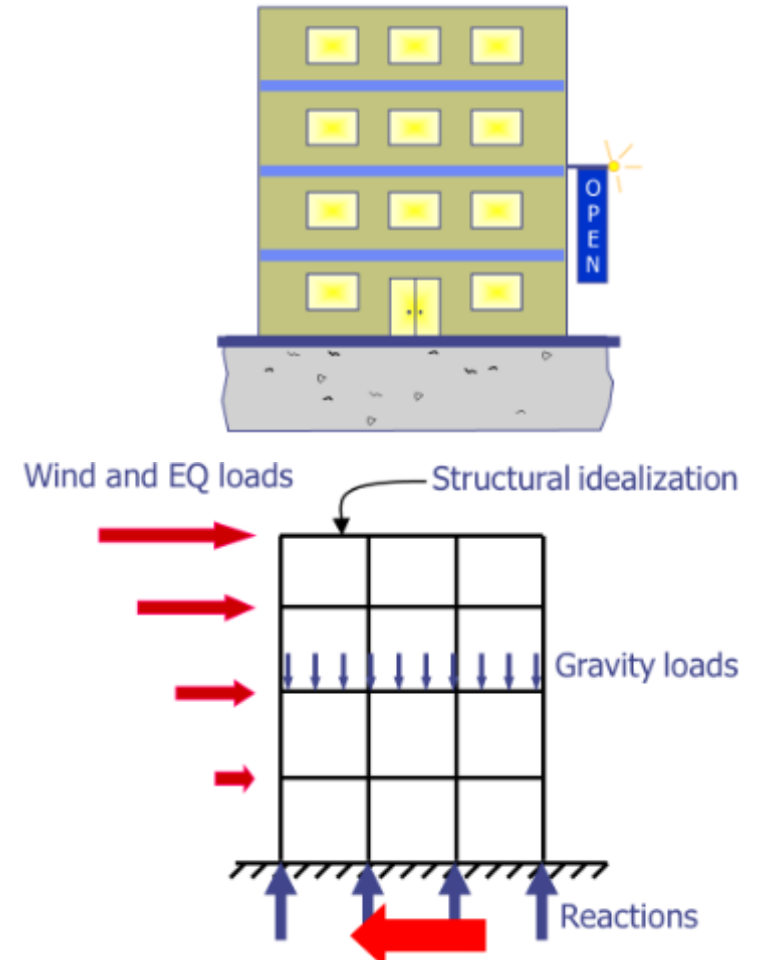
2. Design Loads

3. Structural Design

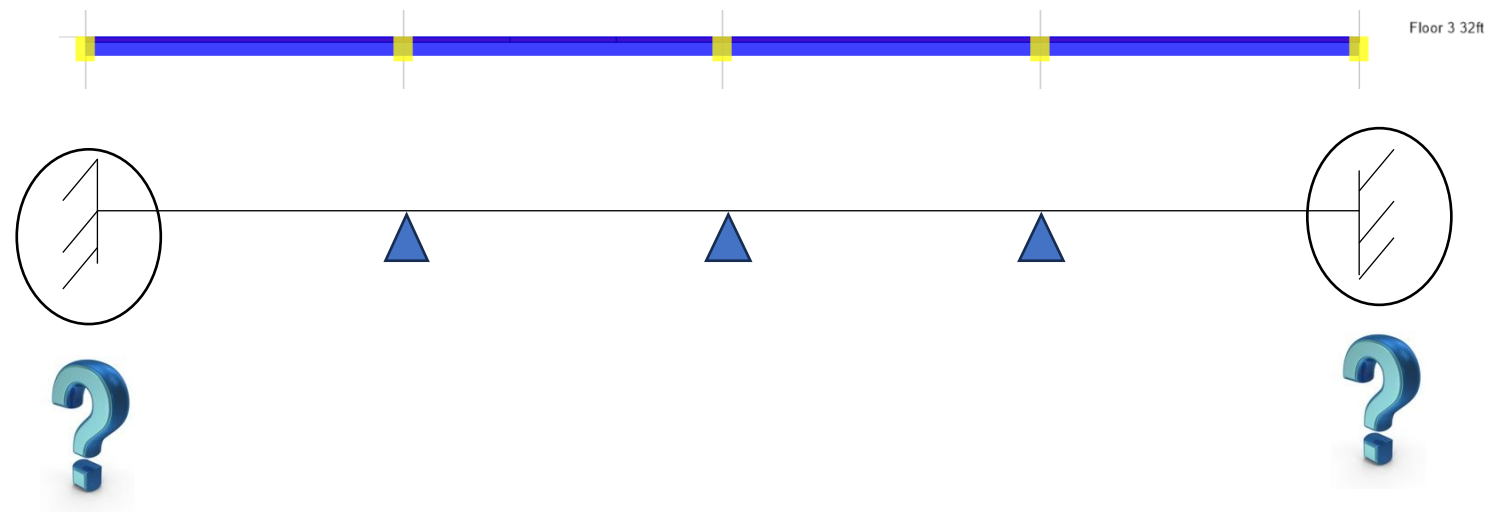


Introduction

1. **Identify a concept for a structural system**
2. Estimate loads
3. Analyze the structure
4. Develop final members and structure proportions
5. Specify the design

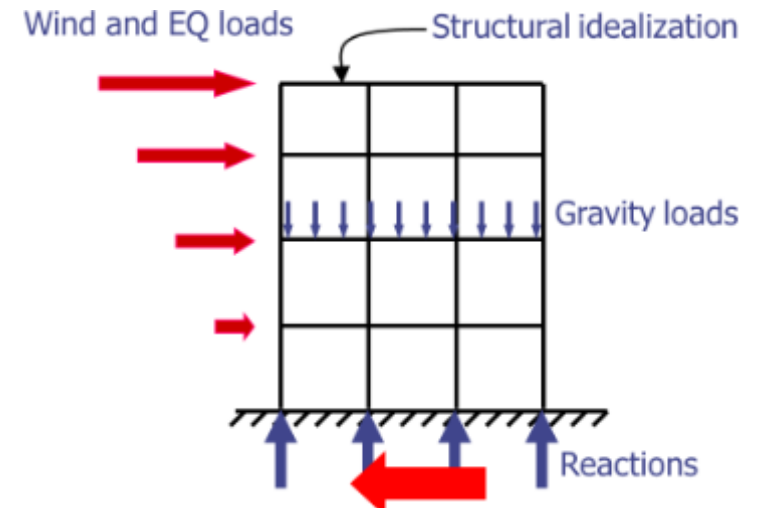
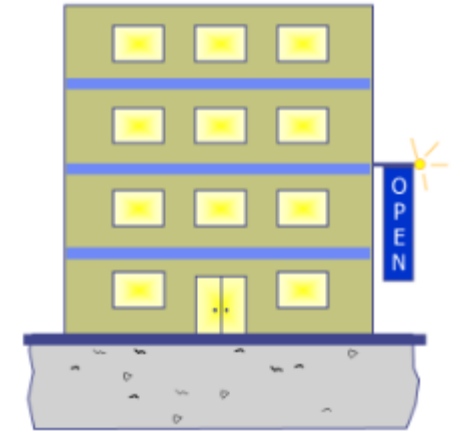


Structural system - Idealized model



Introduction

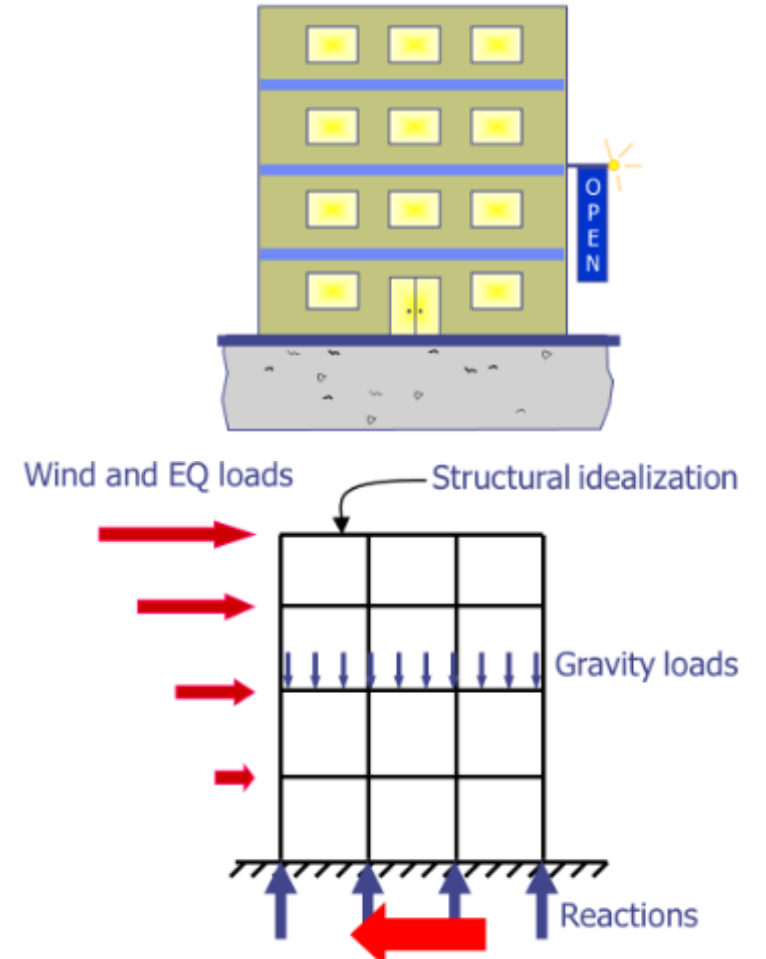
1. Identify a concept for a structural system
- 2. Estimate loads**
3. Analyze the structure
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Introduction

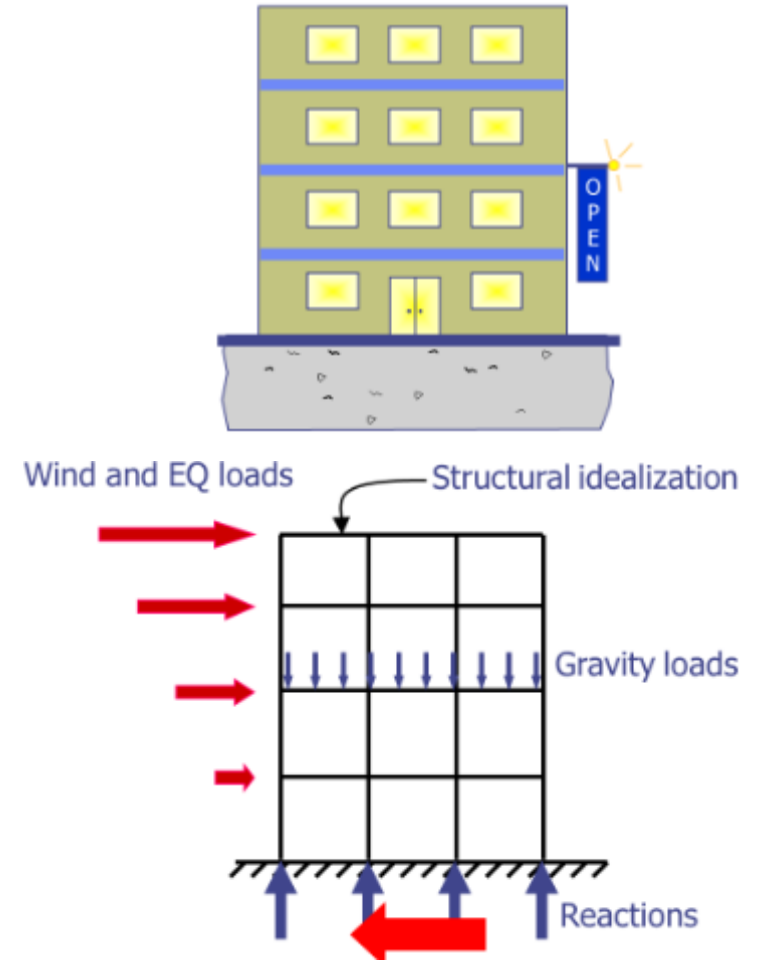
1. Identify a concept for a structural system
2. Estimate loads
- 3. Analyze the structure**
- 4. Develop final members and structure proportions**
- 5. Specify the design**

This material is going to be covered in depth during your classes!!



Contents

1. Introduction
- 2. Design Loads**
3. Structural Design

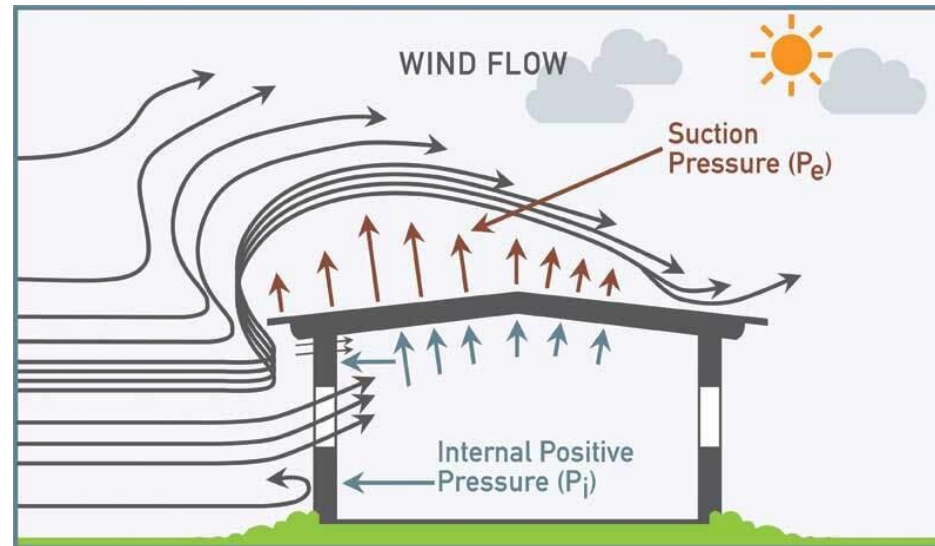
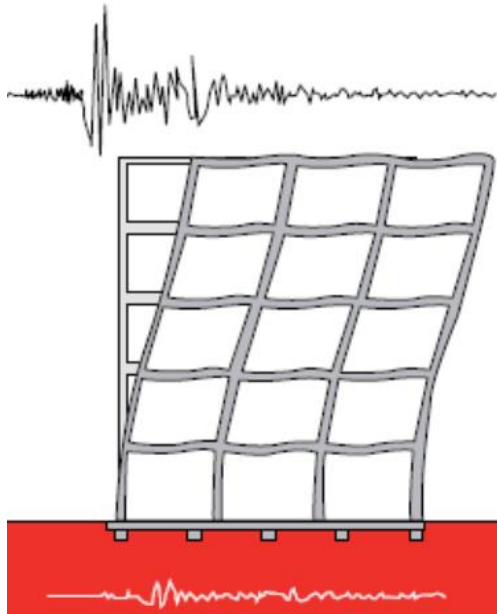


Estimated Loads (Design Loads)

SOURCES

Externally applied forces

- Self-weight
- Equipment
- Live loads
- Earthquake load
- Wind loads, etc.

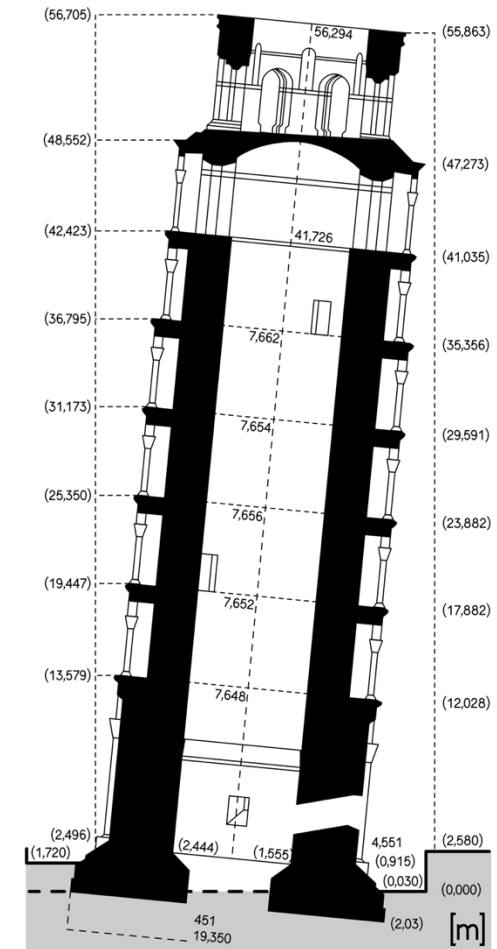


Estimated Loads (Design Loads)

SOURCES

Imposed deformations

- Expansion due to temperature change
- Lack of fit
- Foundation settlement



Estimated Loads (Design Loads)

Load types

Main load types according to ASCE 7

D = Dead load

F = Load due to fluids

Fa = Flood load

H = Load due to lateral earth pressure

L = Live load

Lr = Roof live load

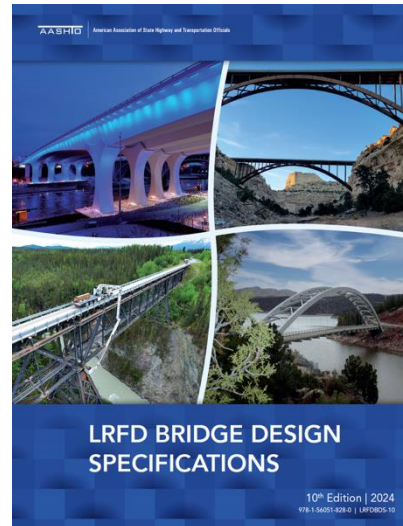
R = Rain load

S = Snow load

T = Self-straining load

W = Wind load

E = Earthquake load



PRESCRIPTIVE



Estimated Loads (Design Loads)

Load types

Main load types according to ASCE 7

D = Dead load (SW + SDL)

F = Load due to fluids

Fa = Flood load

H = Load due to lateral earth pressure

L = Live load

Lr = Roof live load

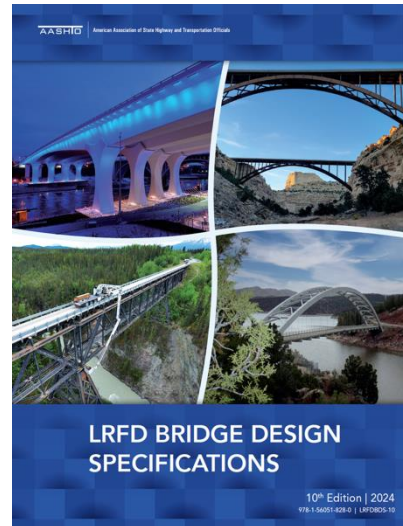
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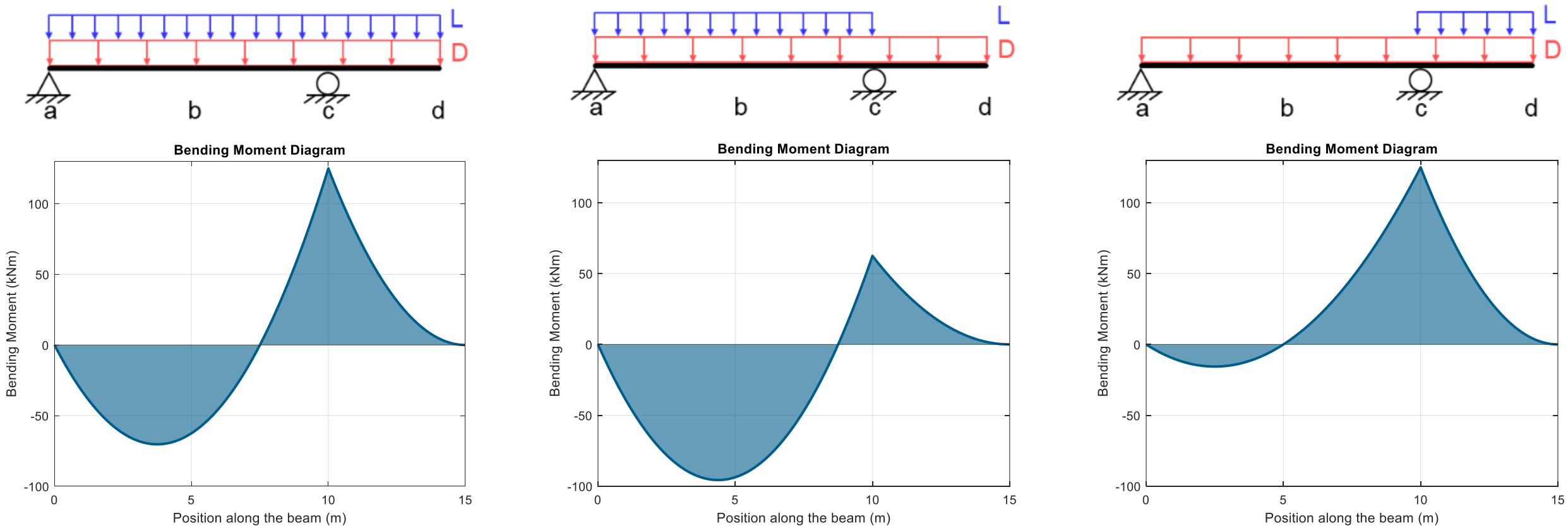
E = Earthquake load



Estimated Loads (Design Loads)

Loading Patterns

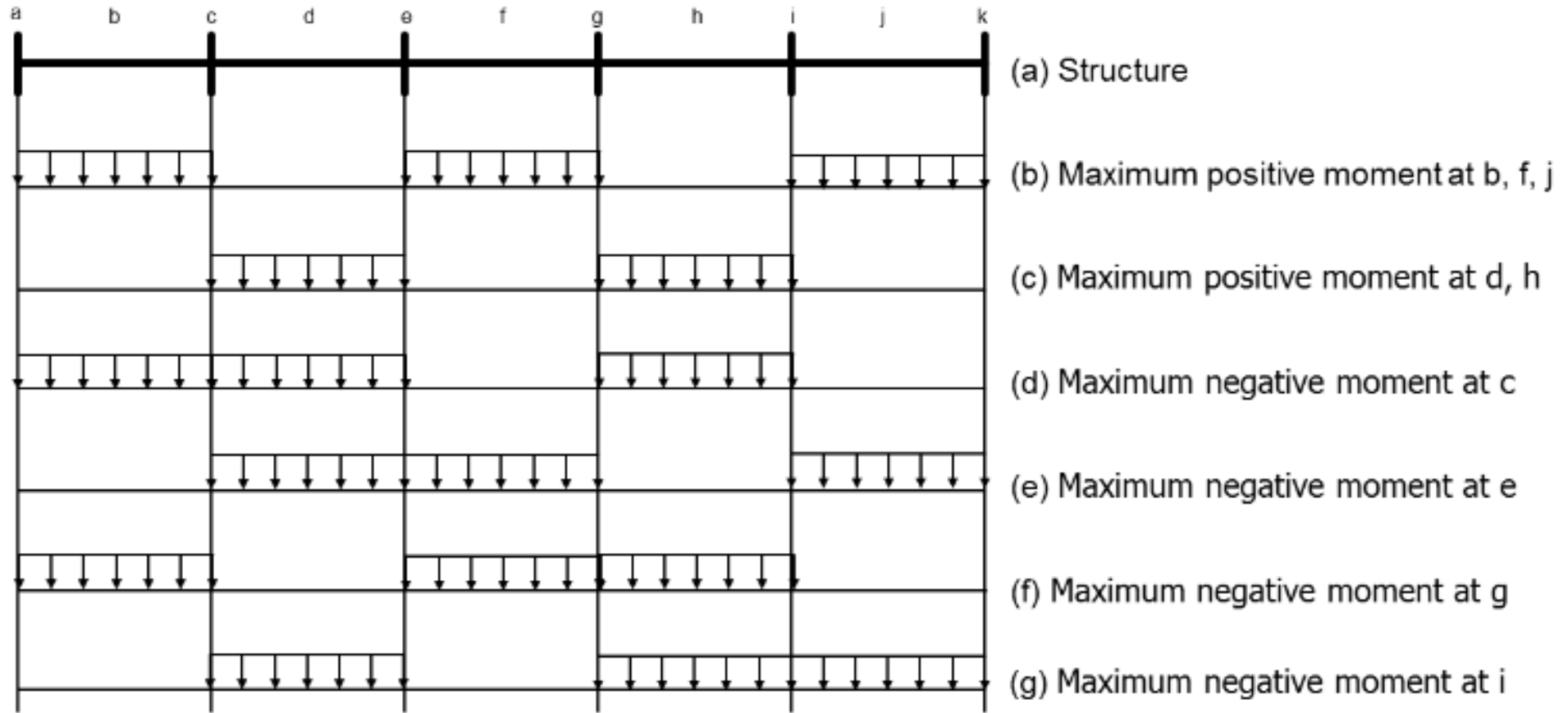
Live loads should be placed in patterns to obtain their maximum effect!



Estimated Loads (Design Loads)

Loading Patterns

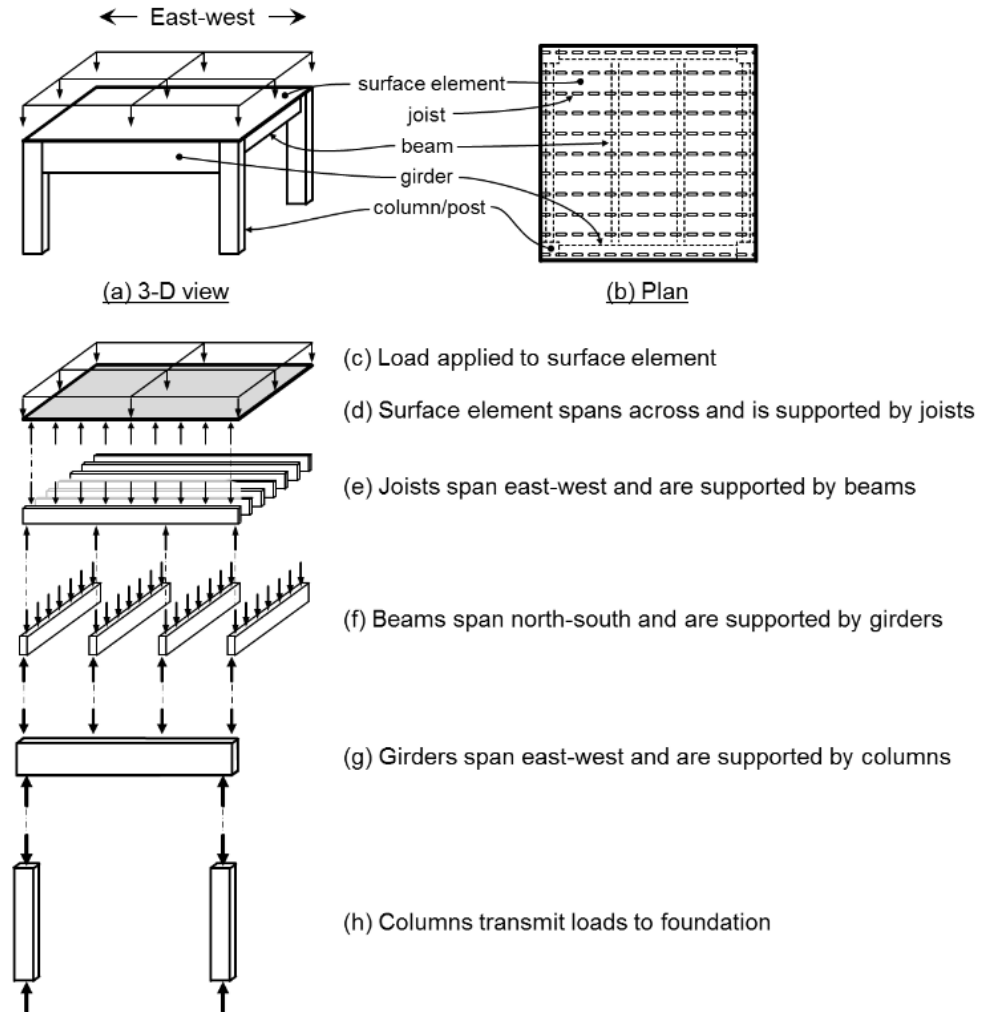
Live loads should be placed in patterns to obtain their maximum effect!



Estimated Loads (Design Loads)

Loading Path

Route through which applied forces travel from the point of application to the final supports or foundations of a structure.

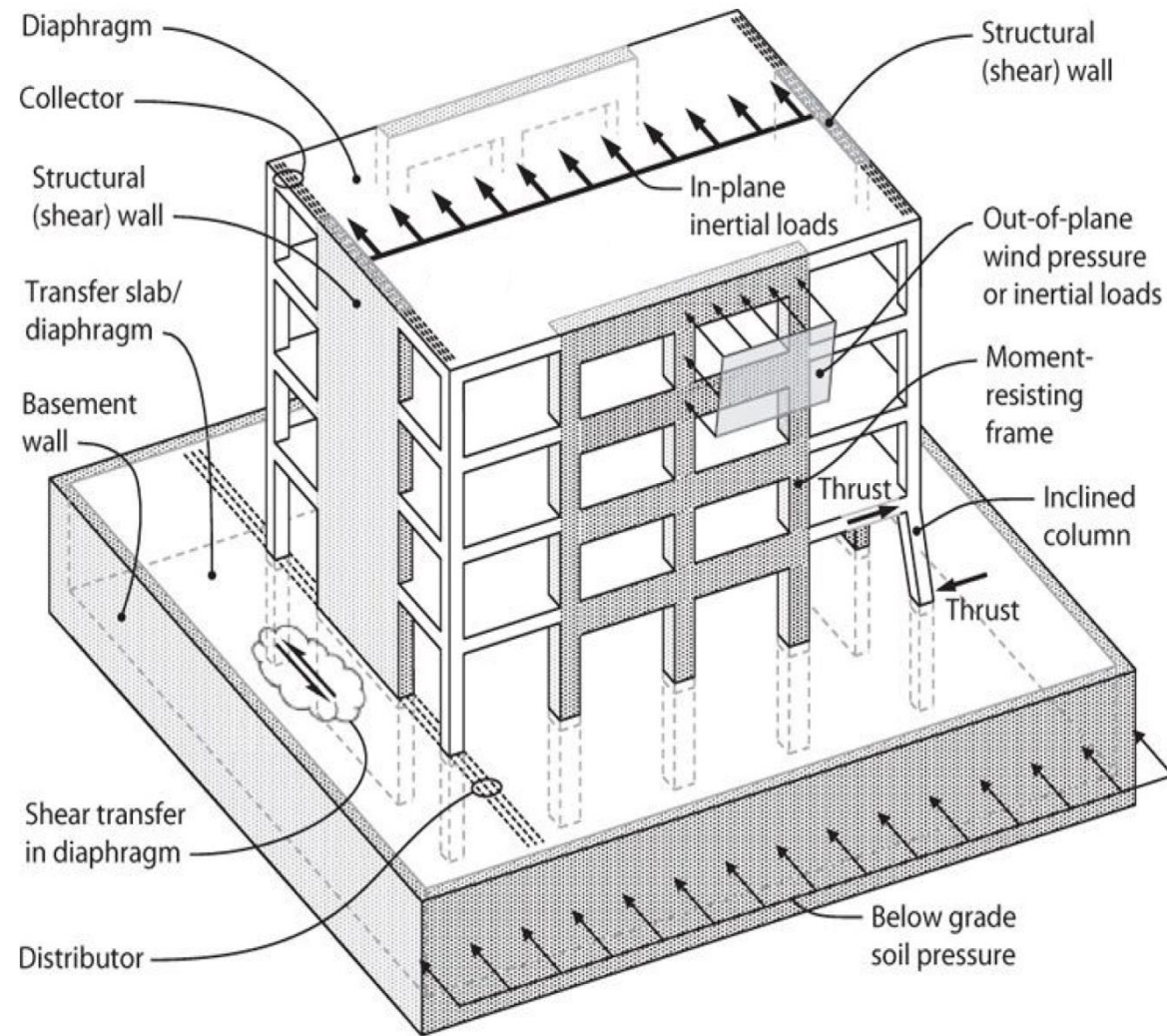


Gravity Loads

Estimated Loads (Design Loads)

Loading Path

Route through which applied forces travel from the point of application to the final supports or foundations of a structure.



Lateral Loads

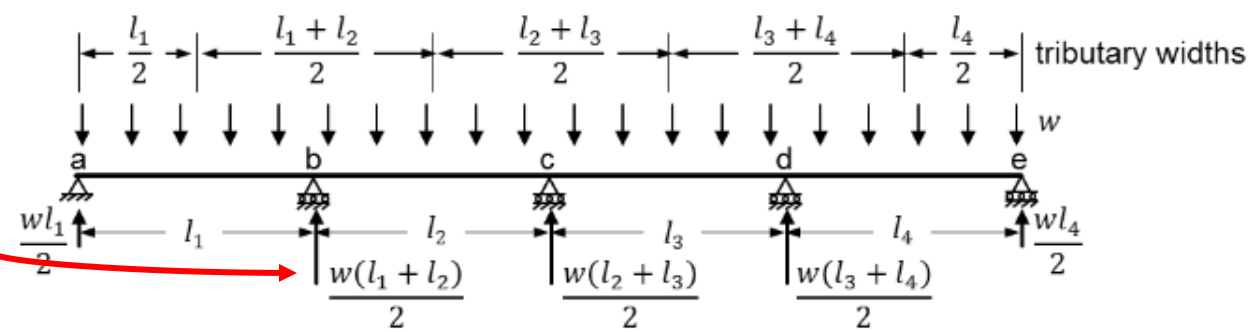
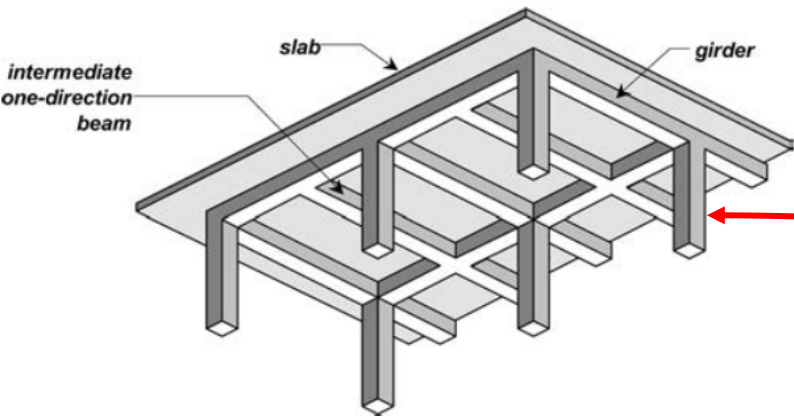
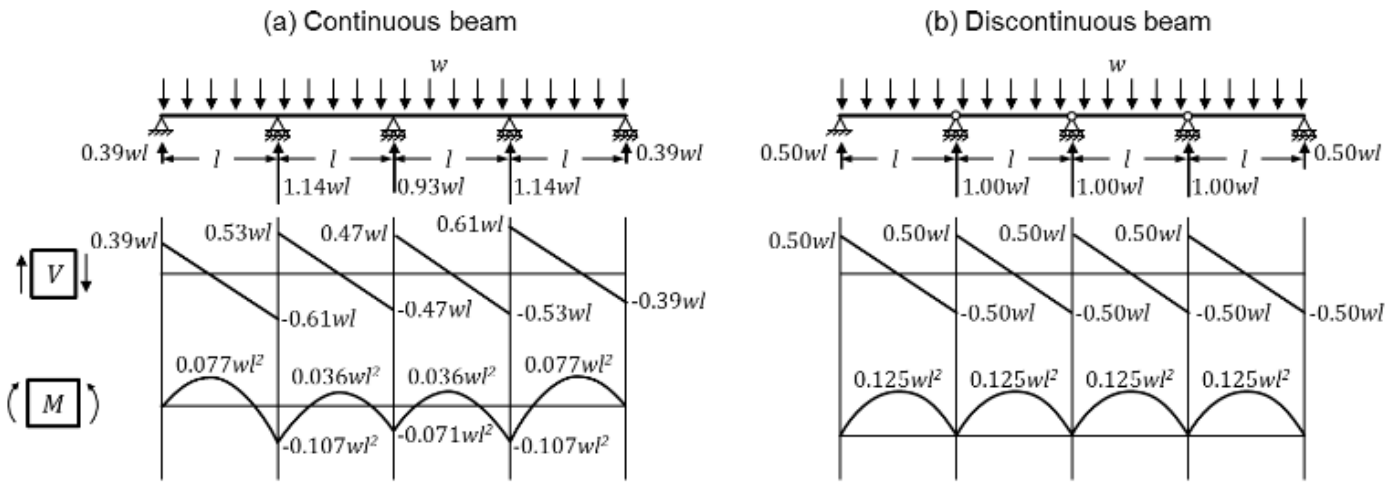
Estimated Loads (Design Loads)

Tributary width

The load transferred to a beam support is equal to the load acting within the tributary width

The tributary width is a width extending halfway to each of the adjacent support

Works well where loads are uniformly distributed

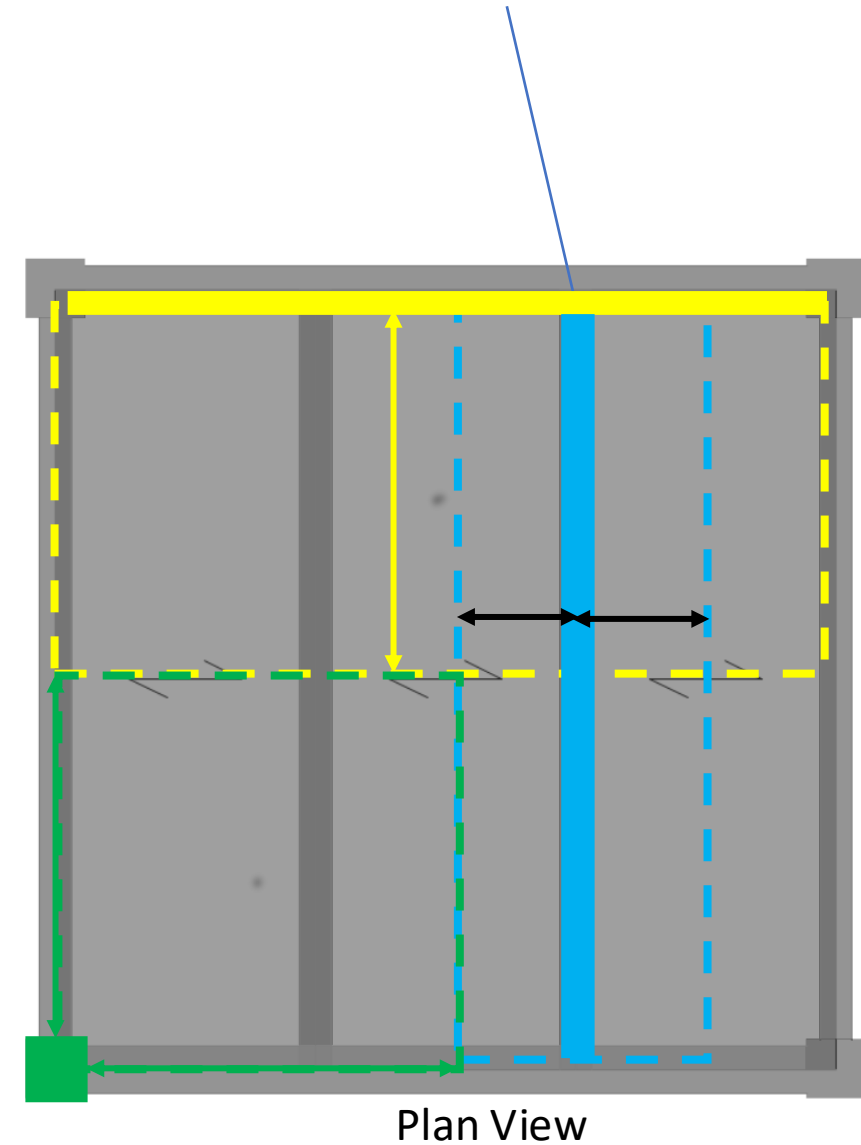
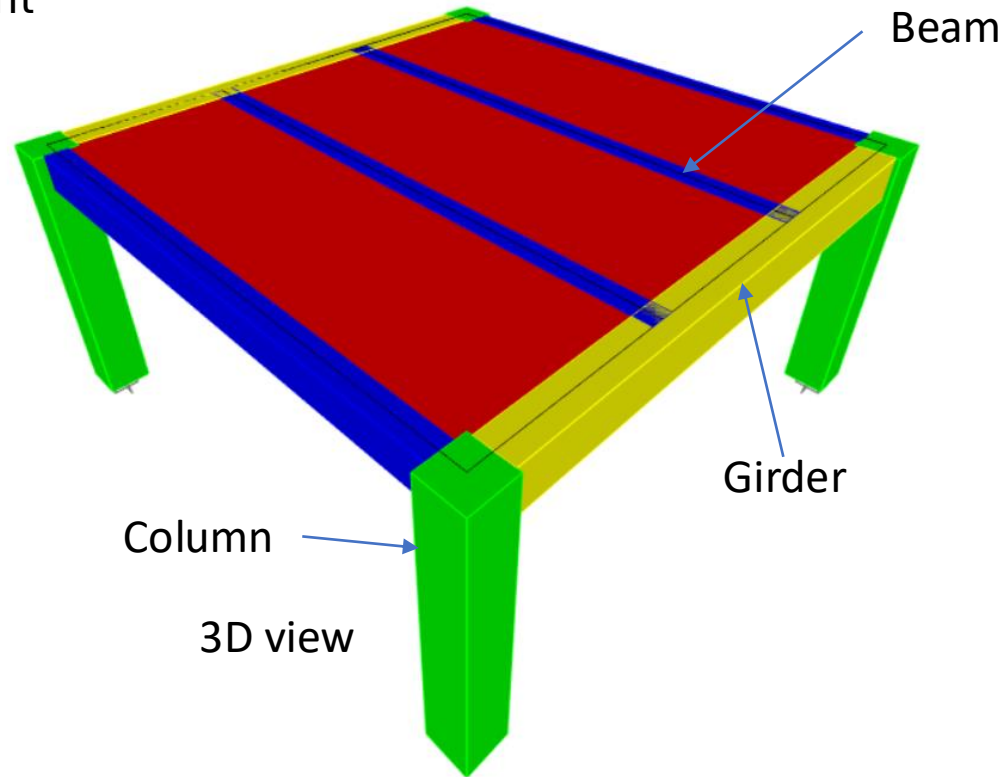


Estimated Loads (Design Loads)

Tributary area

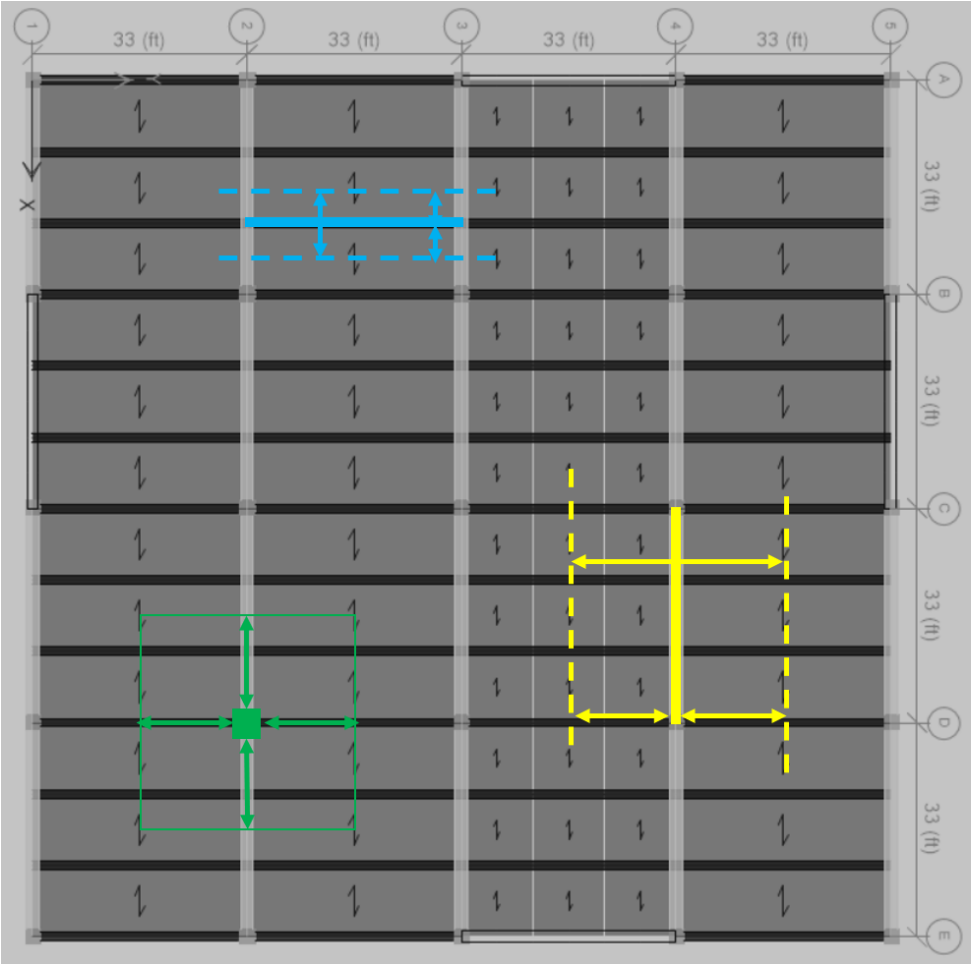
Similar concept to tributary width

Region that contributes load to a specific structural element



Estimated Loads (Design Loads)

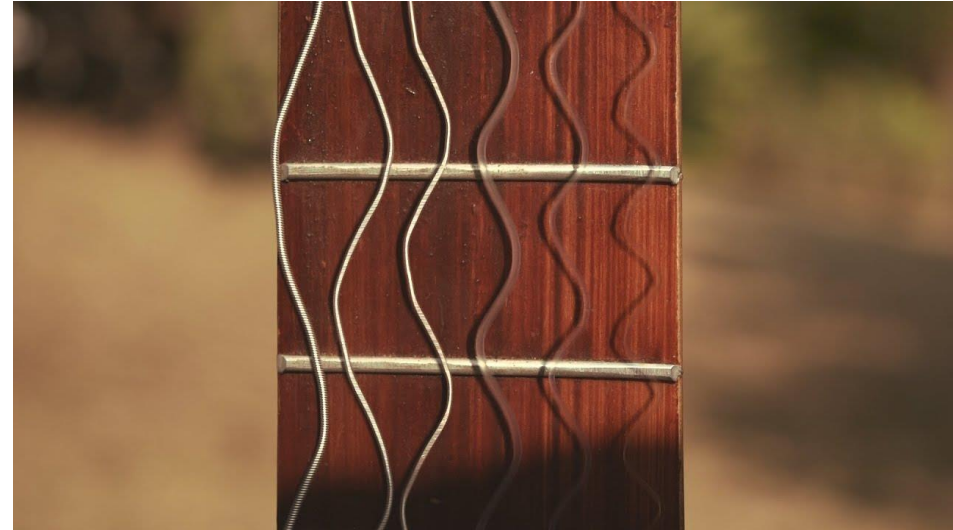
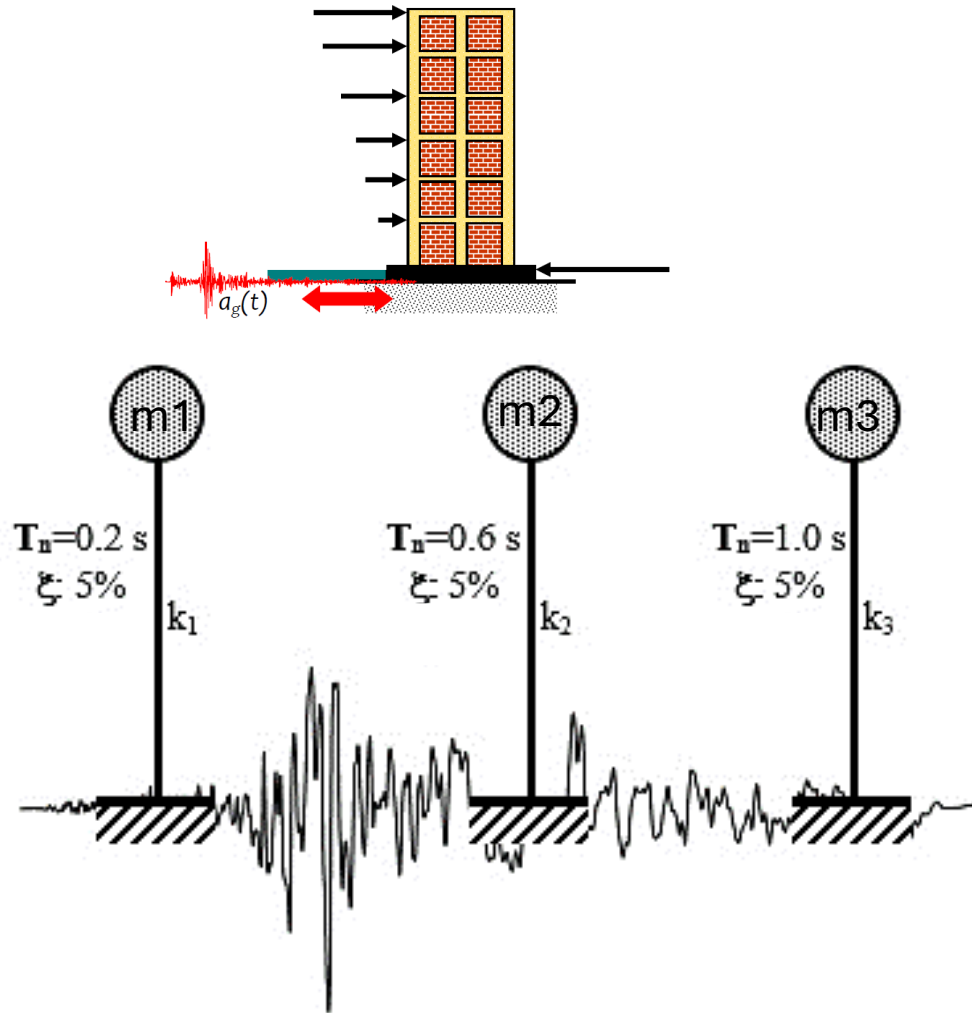
Tributary area



Plan View

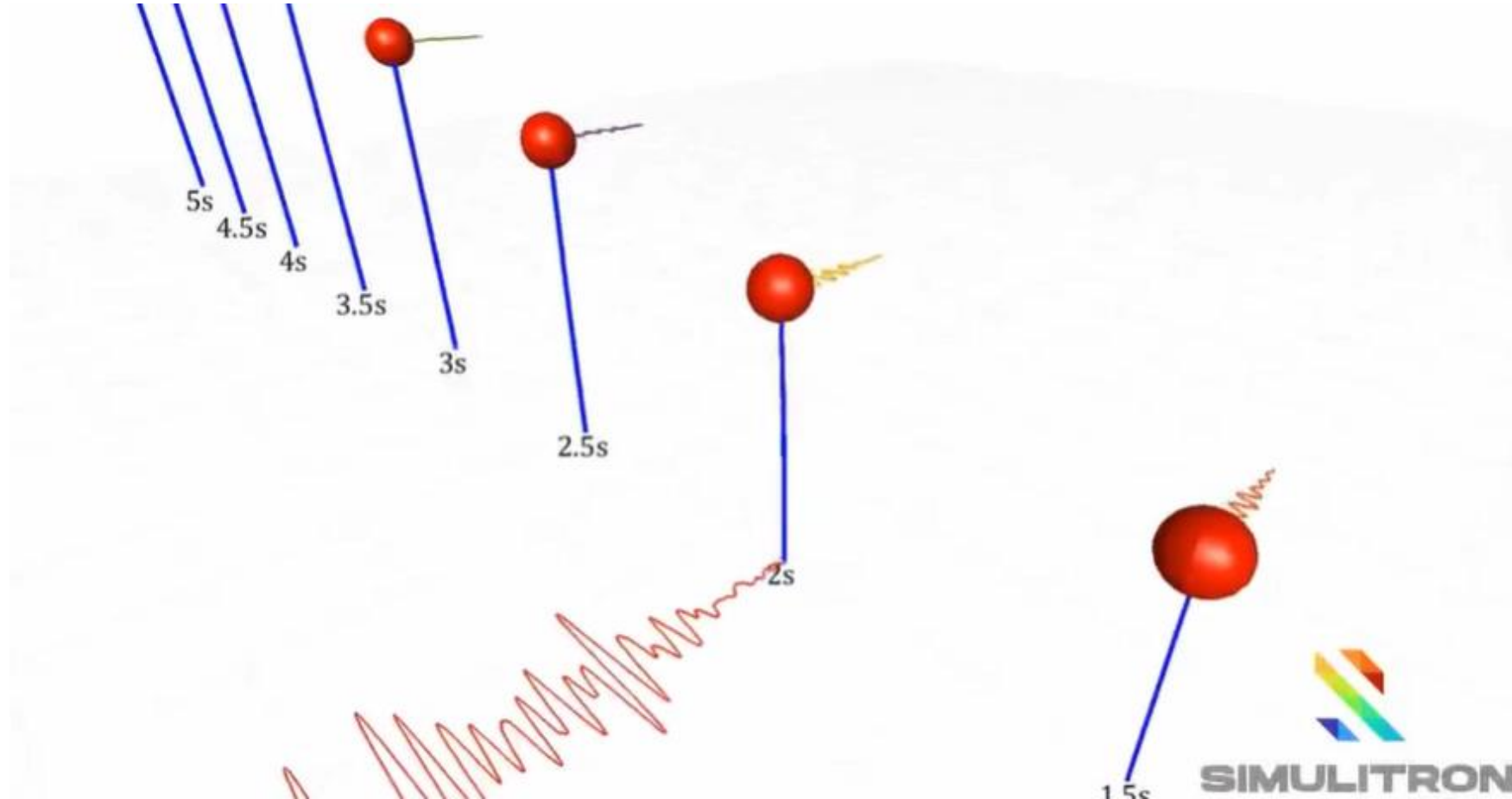
Estimated Loads (Design Loads)

Earthquake Loads



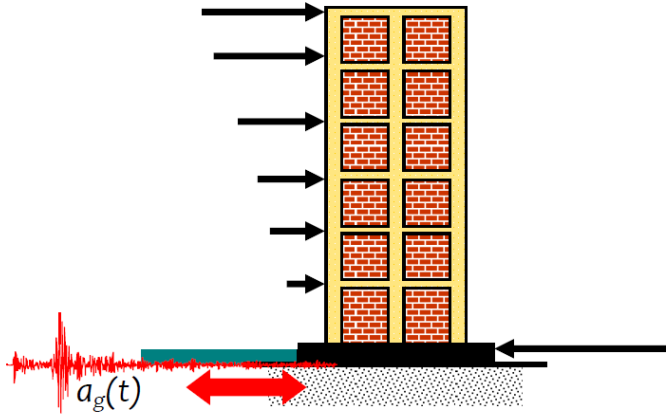
Estimated Loads (Design Loads)

Earthquake Loads



Estimated Loads (Design Loads)

Earthquake Loads



12.8.1 Seismic Base Shear The seismic base shear, V , in a given direction shall be determined in accordance with the following equation:

$$V = C_s W \quad (12.8-1)$$

where

C_s = The seismic response coefficient determined in accordance with Section 12.8.1.1, and

W = The effective seismic weight per Section 12.7.2.

$$\vec{F}_{ext} = m \vec{a}_{com}$$

12.8.3 Vertical Distribution of Seismic Forces The lateral seismic force, F_x (kip or kN), induced at any level shall be determined from the following equations:

$$F_x = C_{vx} V \quad (12.8-12)$$

$$C_{vx} = \frac{w_x h_x^k}{\sum_{i=1}^n w_i h_i^k} \quad (12.8-13)$$

where

C_{vx} = Vertical distribution factor;

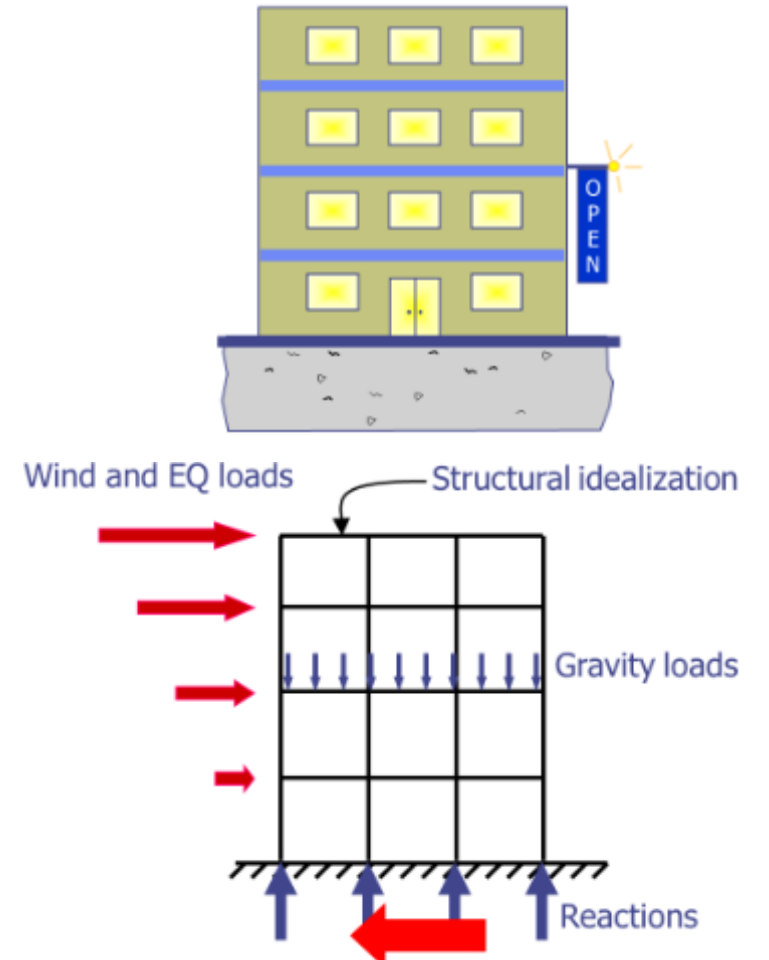
V = Total design lateral force or shear at the base of the structure [kip (kN)];

w_i, w_x = Portion of the total effective seismic weight of the structure (W) located or assigned to level i or x ;

h_i, h_x = Height [ft (m)] from the base to level i or x ; and

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Structural Design

Load and Resistance Factor Design (LRFD)

Method to design ultimate limit state

Load and resistance consider:

Variability and uncertainty in load effects and material properties

Accuracy and variability of nominal strengths

Brittleness of different failure modes

Consequences of failure

$$\phi S_n \geq U$$

ϕ = Strength reduction factor

S_n = Nominal strength

U = **Factored** load effects

**THIS METHOD DOES NOT
COVER CALCULATIONS OR
CONSTRUCTION MISTAKES!**

Structural Design

Load and Resistance Factor Design (LRFD)

Probably you have seen...

$$\phi V_n \geq V_u$$

Nominal shear strength

Shear due to factored loads

$$\phi M_n \geq M_u$$

Nominal moment strength

Moment due to factored loads

$$\phi S_n \geq U$$

ϕ = Strength reduction factor

S_n = Nominal strength

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Structural Design

Load and Resistance Factor Design (LRFD)

Probably you have seen...

$$\phi V_n \geq V_u$$

Nominal shear strength

Shear due to factored loads

$$\phi M_n \geq M_u$$

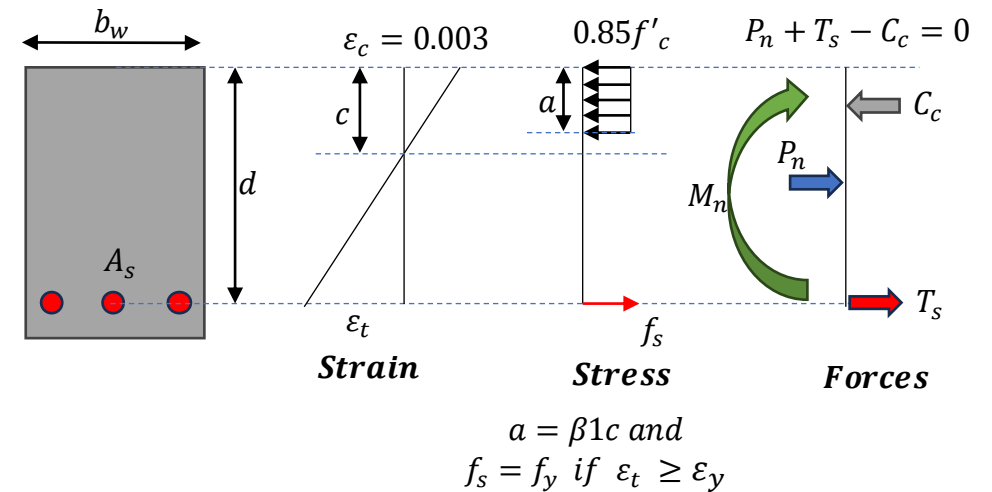
Nominal moment strength

Moment due to factored loads

Nominal strengths:

Strengths that are calculated using methods specified in the building codes.

Defined material properties.



Structural Design

LRFD – ASCE 7 Factored Load Combinations

Basic load combinations

- 1) $1.4D$
- 2) $1.2D + 1.6L + 0.5(L_r \text{ or } S \text{ or } R)$
- 3) $1.2D + 1.6(L_r \text{ or } S \text{ or } R) + (\alpha_L L \text{ or } 0.5W)$
- 4) $1.2D + 1.0W + \alpha_L L + 0.5(L_r \text{ or } S \text{ or } R)$
- 5) $1.2D + 0.2SDS D + 1.0E + \alpha_L L + 0.2S$
- 6) $0.9D + 1.0W$
- 7) $0.9D - 0.2SDS D + 1.0E$

$$\phi S_n \geq U$$

ϕ = Strength reduction factor

S_n = Nominal strength

U = Factored load effects

Structural Design

LRFD – ASCE 7 Factored Load Combinations

Basic load combinations

- 1) $1.4D$
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- 3) $1.2D + 1.6(L_r \text{ or } S \text{ or } R) + (\alpha_L L \text{ or } 0.5W)$
- 4) $1.2D + 1.0W + \alpha_L L + 0.5(L_r \text{ or } S \text{ or } R)$
- 5) $1.2D + 0.2SDS D + 1.0E + \alpha_L L + 0.2S$
- 6) $0.9D + 1.0W$
- 7) $0.9D - 0.2SDS D + 1.0E$

Recall

D = Dead load

F = Load due to fluids

Fa = Flood load

H = Load due to lateral earth pressure

L = Live load

Lr = Roof live load

R = Rain load

S = Snow load

T = Self-straining load

W = Wind load

E = Earthquake load

Structural Design

LRFD – ASCE 7 Factored Load Combinations

Load cases

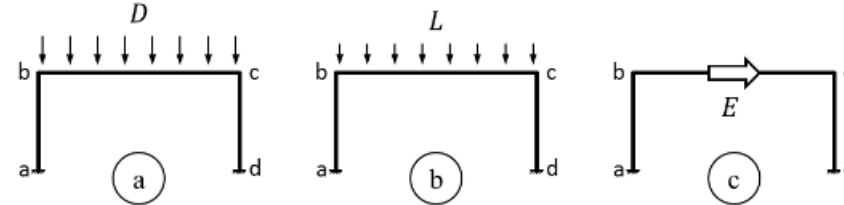
D , L , and E

Analyze each load case separately

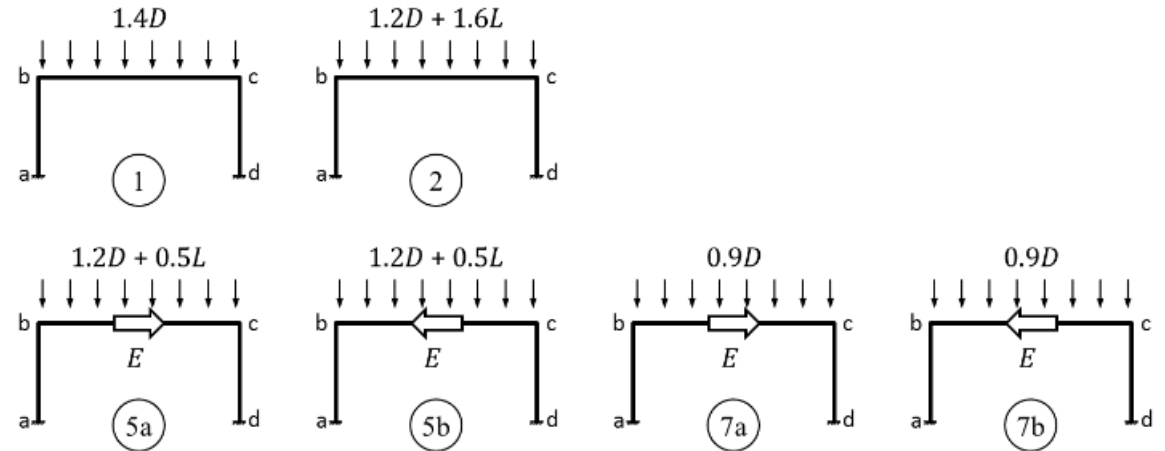
Use linear superposition to compute load combinations

$1.4D$, etc...

(i) Load cases



(ii) Load combinations



Structural Design

Example

Example 6.2 (Single-bay, single-story, frame). A weightless, one-bay, one-story frame has configuration and loading shown in Fig. 6.12. Dead load D is 3 klf (44 kN/m), live load L is 1.8 klf (26 kN/m), and earthquake load E is 45 kips (200 kN). Use the LRFD method to determine the required beam moment strengths at the faces of the beams (Sections 1 and 2).

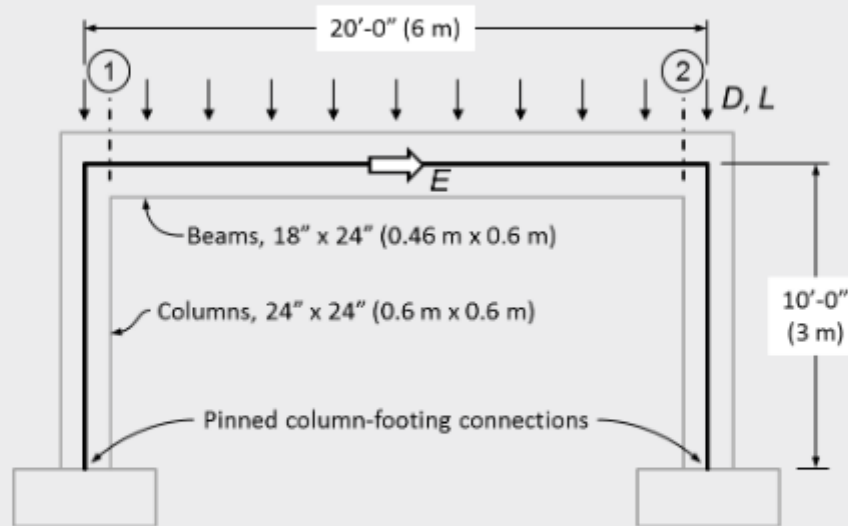
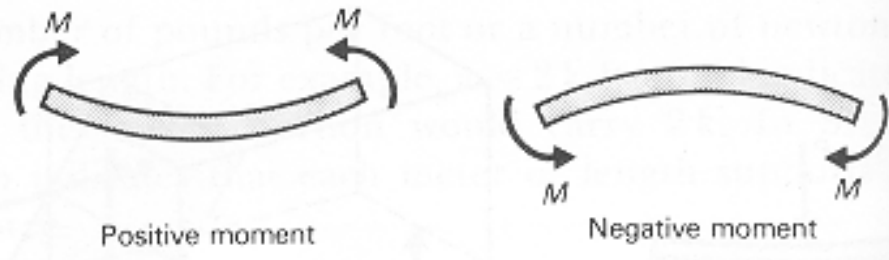


Figure 6.12: Single-bay, single-story, frame.

Solution: The load cases and load combinations are shown in Fig. 6.11. The structure is modeled using flexural stiffness equal to $0.3EI_g$ for beams and columns and analyzed for the load cases using computer software for structural analysis. The results of the load cases are then combined using the load combinations. Calculated moments at sections 1 and 2 are tabulated below.

Table 6.1: Load cases and combinations.

Moments, k-ft (kN-m)		
Load case	Section 1	Section 2
D	-51.6 (-70.2)	-51.6 (-70.2)
L	-31.0 (-42.1)	-31.0 (-42.1)
E	203 (275)	-203 (-275)
Load Combination		
$1.4D$	-72.2 (-98.2)	-72.2 (-98.2)
$1.2D + 1.6L$	-111 (-152)	-111 (-152)
$1.2D + 0.5L + E$	125 (170)	-280 (-381)
$1.2D + 0.5L - E$	-280 (-381)	125 (170)
$0.9D + E$	156 (212)	-249 (-339)
$0.9D - E$	-249 (-339)	156 (212)
Minimum	-280 (-381)	-280 (-381)
Maximum	156 (212)	156 (212)

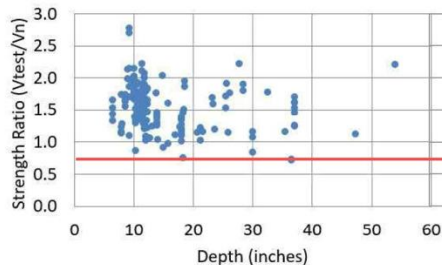


Structural Design

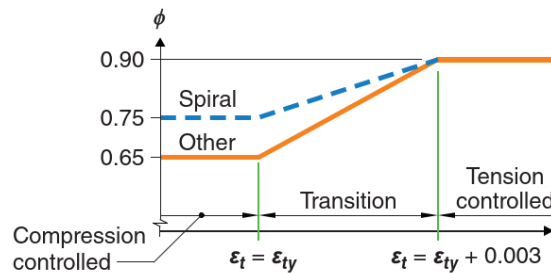
LRFD – Resistance Factors

- $\phi < 1.0$
- Accounts for under-strength due to material variations or dimensions
- Allows inaccuracies in design equations
- Reflect member ductility and reliability
- Reflect the member's structural importance
- For concrete elements:

$$\phi_v = 0.75$$



$$\phi_f = 0.90$$



$$\phi S_n \geq U$$

ϕ = Strength reduction factor

S_n = Nominal strength

U = Factored load effects

**THIS METHOD DOES NOT
COVER CALCULATIONS OR
CONSTRUCTION MISTAKES!**

Structural Design

Example

Example 6.3 (Required nominal moment strengths). Determine the required nominal moment strengths of the beam at section 1 considering the loading of Example 6.2. Assume the strength reduction factor is $\phi = 0.9$ for beam moment strength.

Minimum	-280 (-381)	-280 (-381)
Maximum	156 (212)	156 (212)

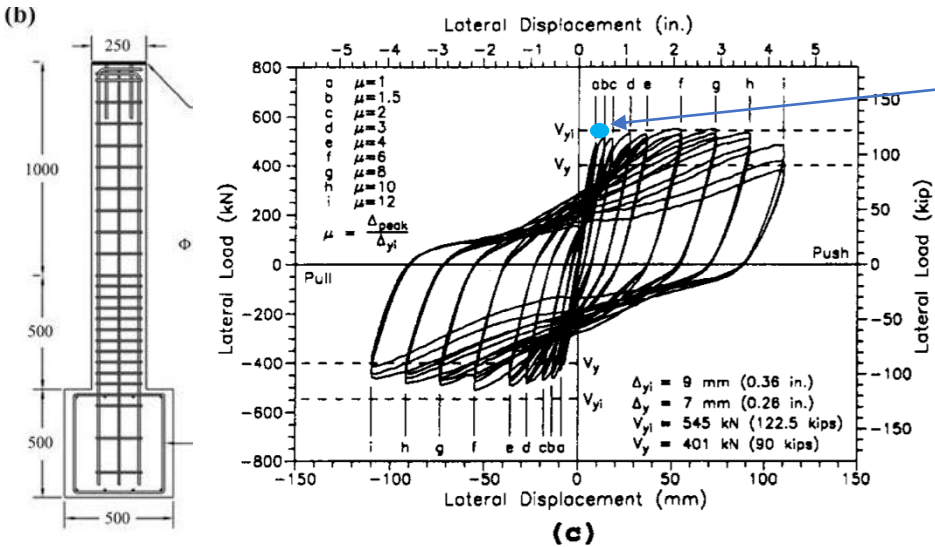
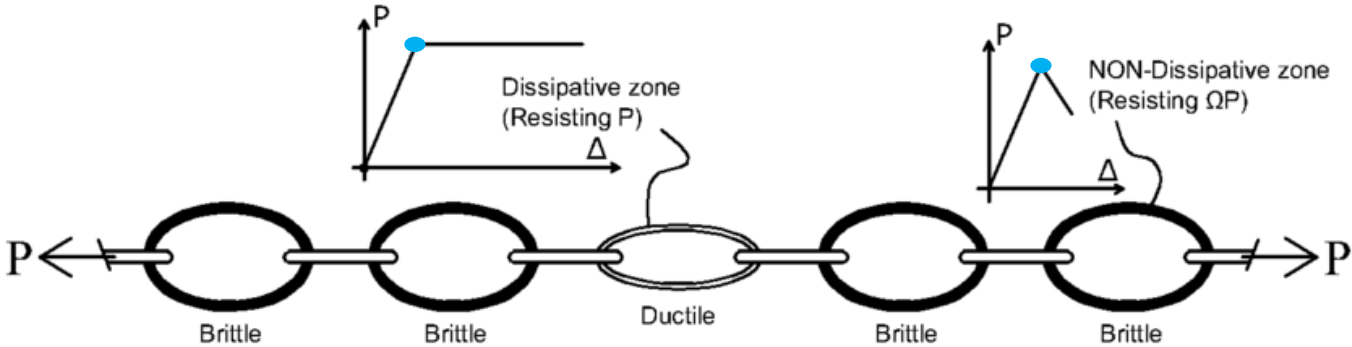
Solution: From Example 6.2, the required moment strengths are $M_u = -280$ k-ft and $+156$ k-ft. Thus, the required nominal moment strengths are $M_n = M_u/\phi = -311$ k-ft and $+173$ k-ft. The beams would need to be designed to provide at least these nominal strengths.

$$M_{n,top} = A_{s,top}f_y \left(d - \frac{a}{2} \right) \geq 311 \text{ kip} \cdot \text{ft} = M_{u,top} \quad \text{5\#8 Top and}$$

$$M_{n,bott} = A_{s,bott}f_y \left(d - \frac{a}{2} \right) \geq 173 \text{ kip} \cdot \text{ft} = M_{u,bott} \quad \text{3\#8 Bottom}$$

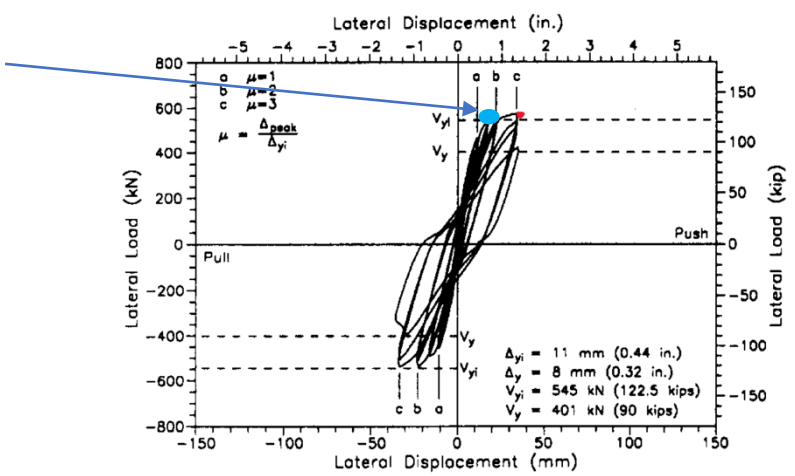
Structural Design

Capacity Design

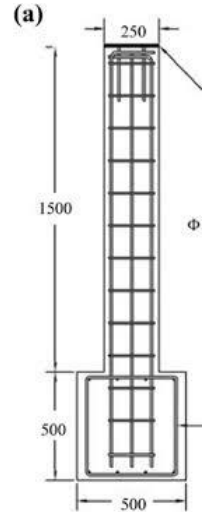


Flexure-controlled

Yielding
occured

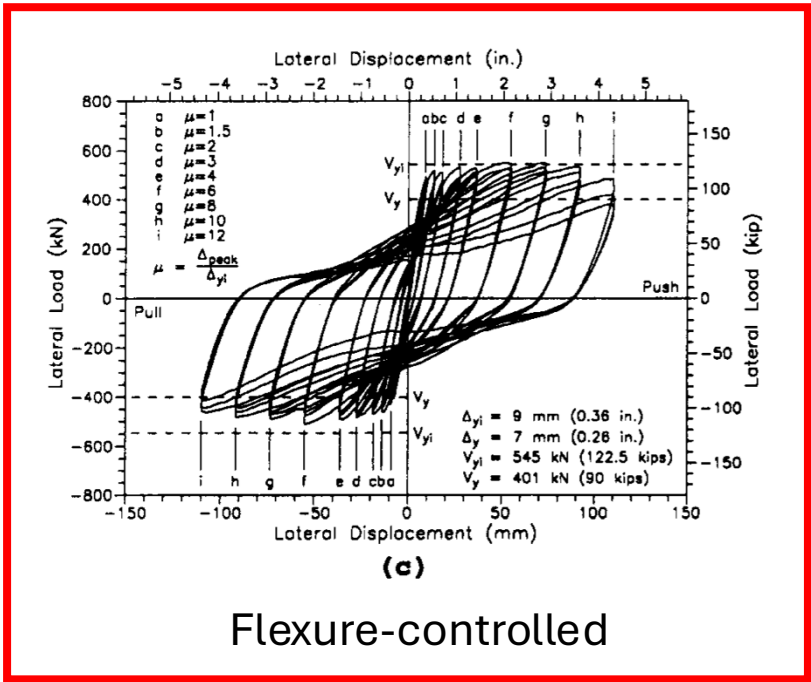
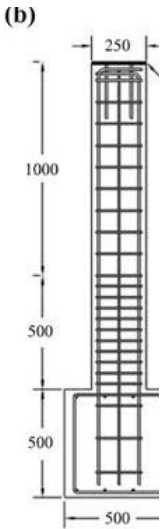
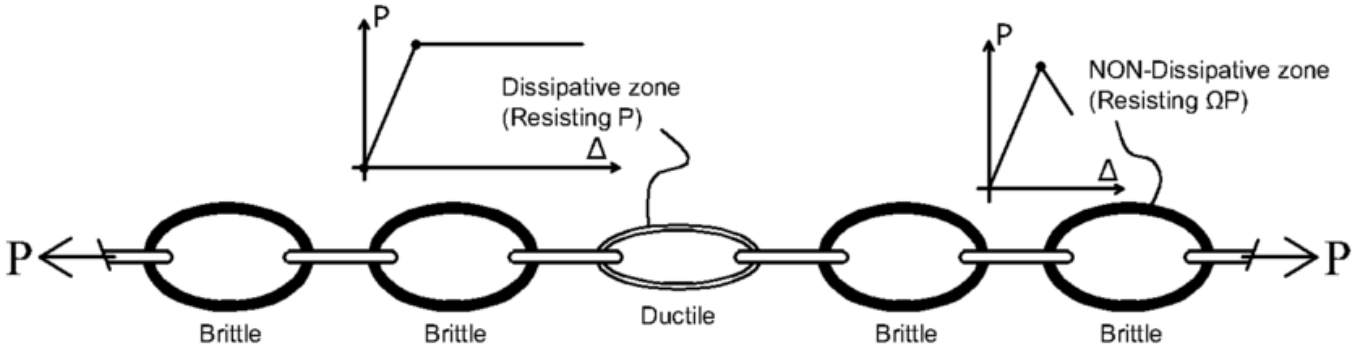


Shear-controlled

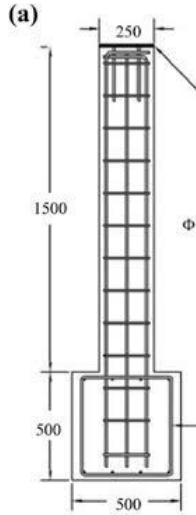
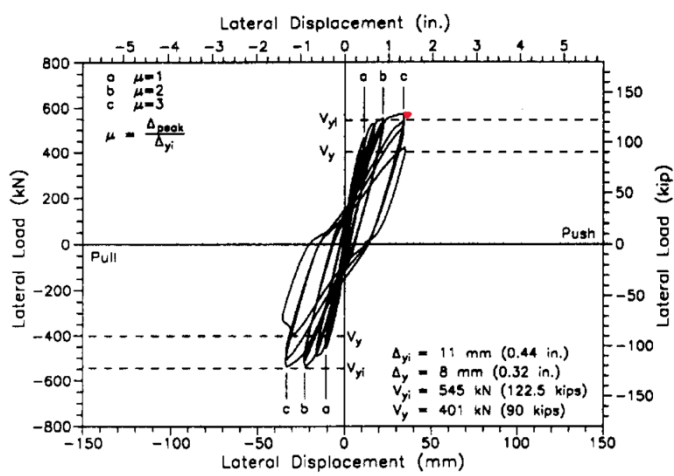


Structural Design

Capacity Design



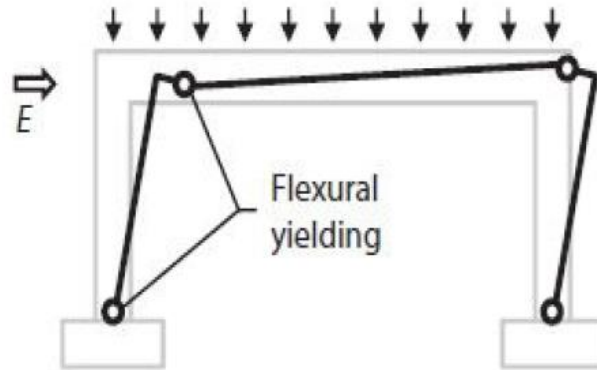
MAKE SURE THIS
HAPPENS



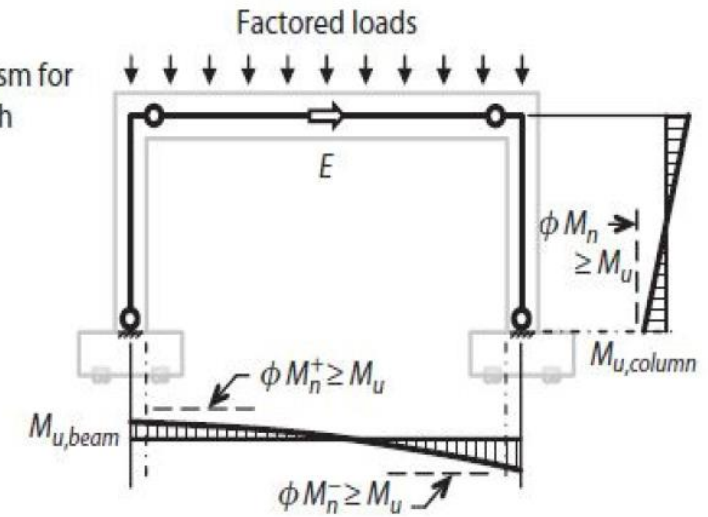
Structural Design

Capacity Design

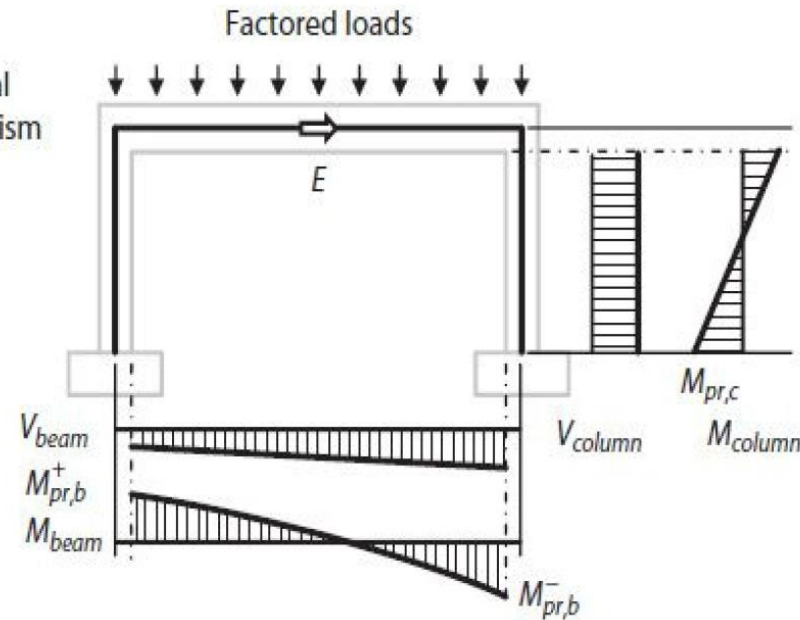
(1) Select yielding mechanism



(2) Design mechanism for required strength

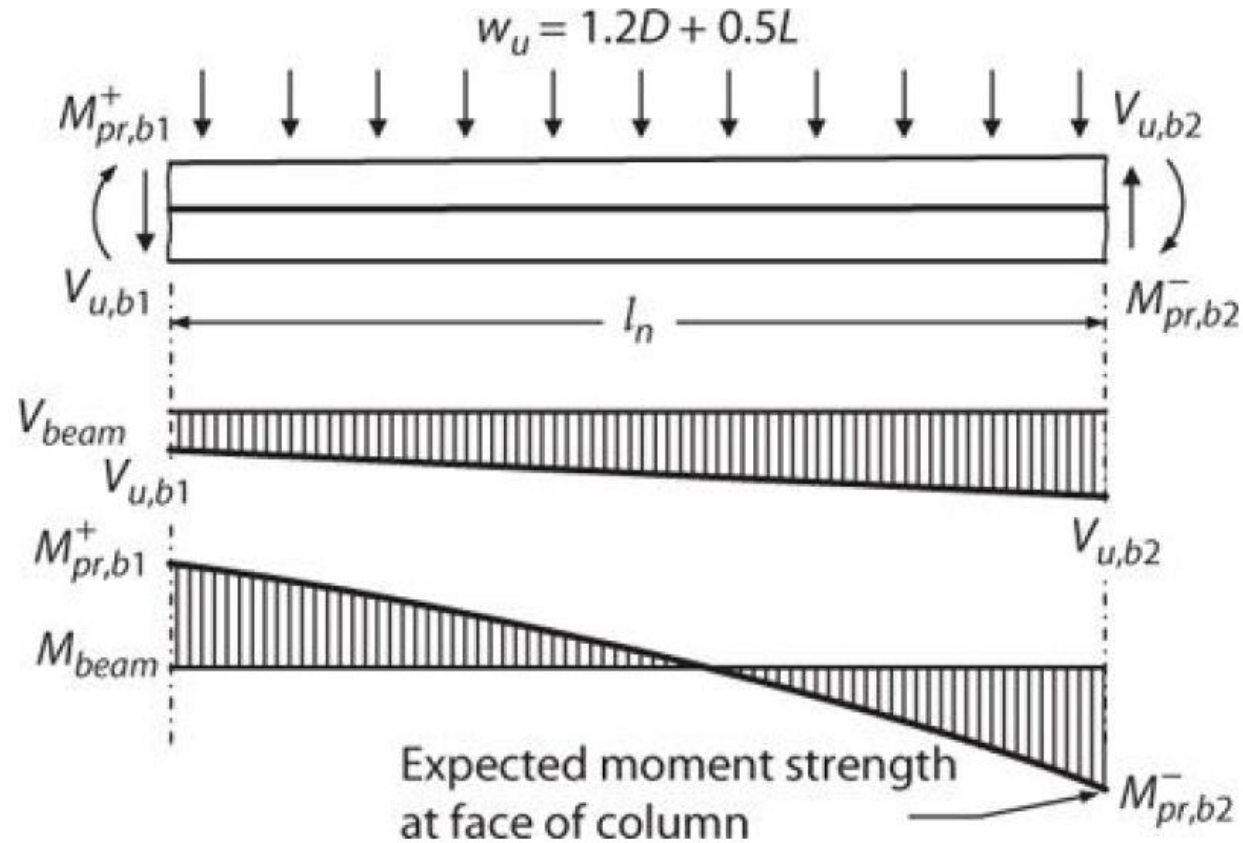


(3) Determine internal forces for mechanism



Structural Design

Capacity Design



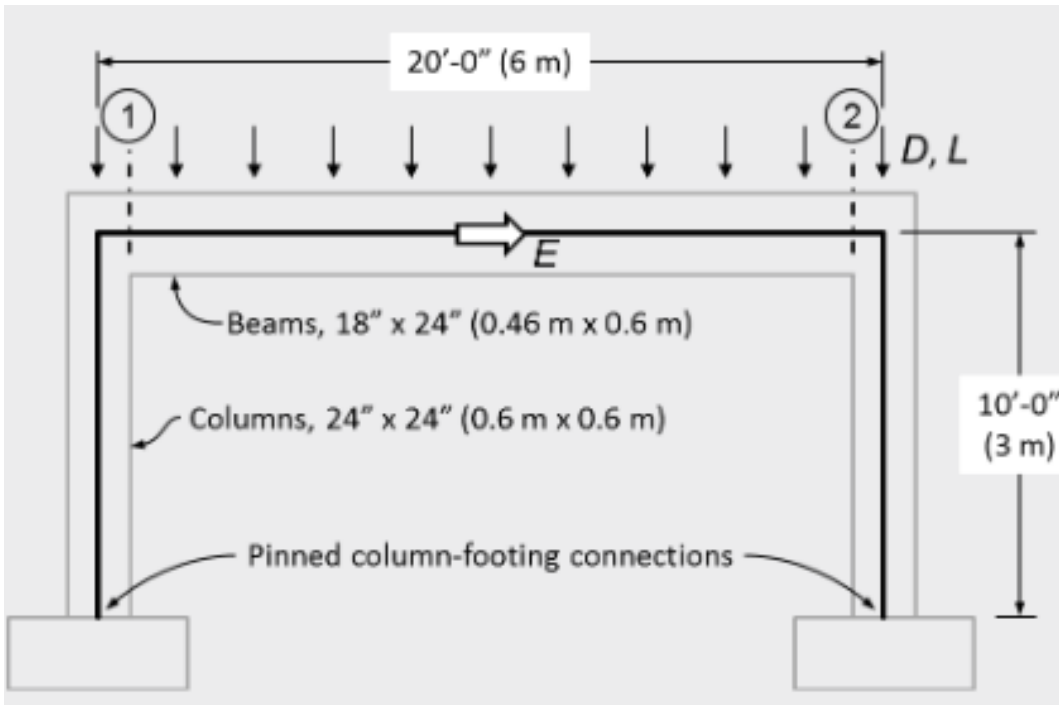
$$V_{u,b1} = \frac{M_{pr,b1}^+ + M_{pr,b2}^-}{l_n} - \frac{w_u l_n}{2}$$

$$V_{u,b2} = \frac{M_{pr,b1}^+ + M_{pr,b2}^-}{l_n} + \frac{w_u l_n}{2}$$

$$\phi V_n \geq V_u$$

Structural Design

EXAMPLE : Find the design shear force following a Capacity Design approach



$$M_{pr,top} = A_{s,top} \mathbf{1.25} f_y \left(d - \frac{a_{pr,top}}{2} \right) = 458 \text{ kip} \cdot \text{ft}$$

$$M_{pr,bott} = A_{s,bott} \mathbf{1.25} f_y \left(d - \frac{a_{pr,bott}}{2} \right) = 289 \text{ kip} \cdot \text{ft}$$

$$a_{pr,top} = \frac{A_{s,top} \mathbf{1.25} f_y}{0.85 f' c b_w} = \frac{5(0.79 in^2) \mathbf{1.25} (60 ksi)}{0.85 (4 ksi) (18 in)} = 4.84 in$$

Structural Design

EXAMPLE Capacity Design

$$M_{pr,top} = M_{pr(-)} = 458 \text{ kip} \cdot \text{ft}$$

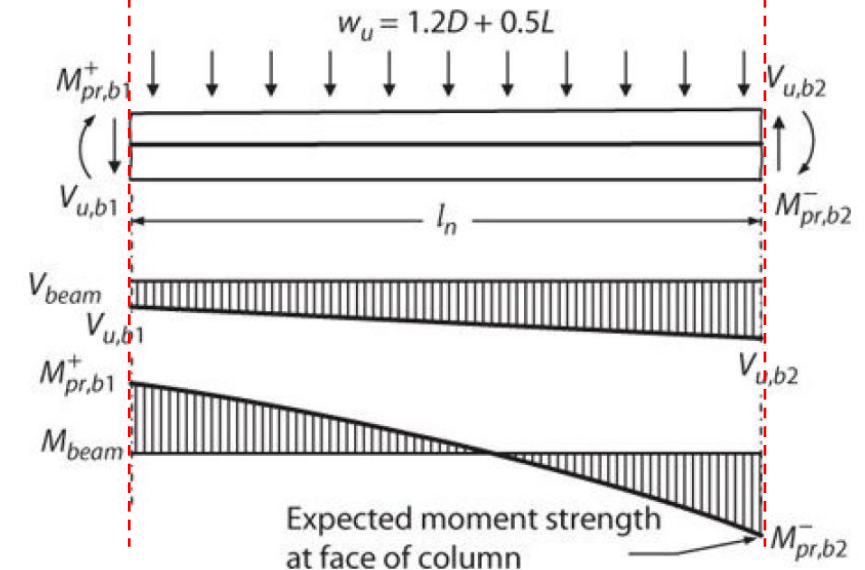
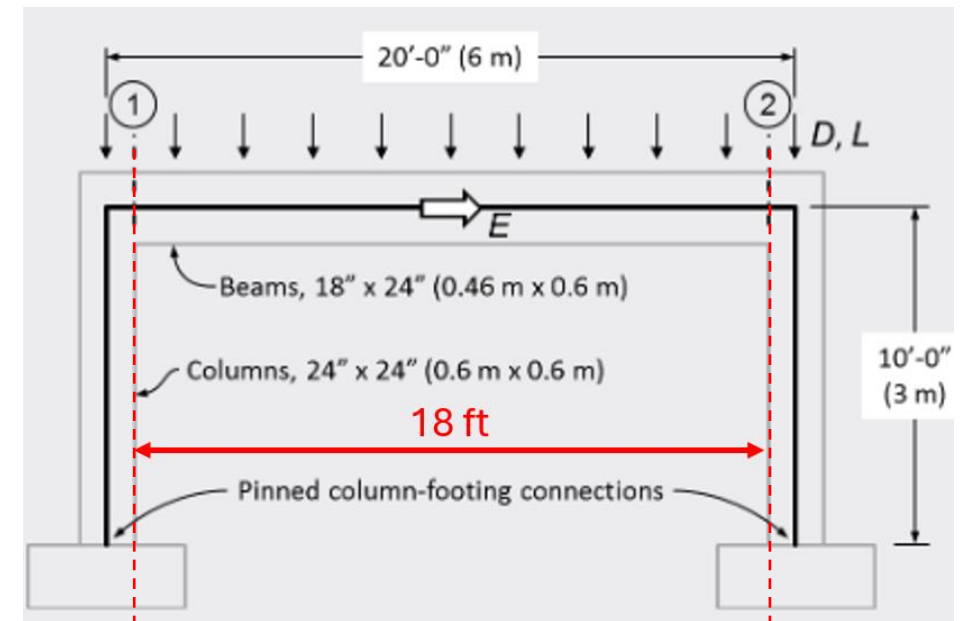
$$M_{pr,bott} = M_{pr(+)} = 289 \text{ kip} \cdot \text{ft}$$

$$w_u = 4.5 \text{ klf}$$

Example 6.2 (Single-bay, single-story, frame). A weightless, one-bay, one-story frame has configuration and loading shown in Fig. 6.12. Dead load D is 3 klf (44 kN/m), live load L is 1.8 klf (26 kN/m), and earthquake load E is 45 kips (200 kN). Use the LRFD method to determine the required beam moment strengths at the faces of the beams (Sections 1 and 2).

$$V_{u,b1} = \frac{M_{pr(+)} + M_{pr(-)}}{l_n} - \frac{w_u l_n}{2} = 1 \text{ kip}$$

$$V_{u,b2} = \frac{M_{pr(+)} + M_{pr(-)}}{l_n} + \frac{w_u l_n}{2} = 82 \text{ kip}$$



Relevant Design Courses in SEMM

CE 244. Reinforced Concrete Structures

CE 247. Design of Steel and Composite Structures

CE 227. Earthquake-Resistant Design

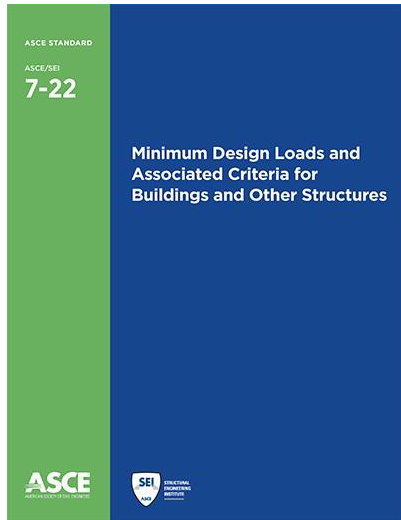
CE 124. Structural Design in Timber

CE 122. Design of Steel Structures

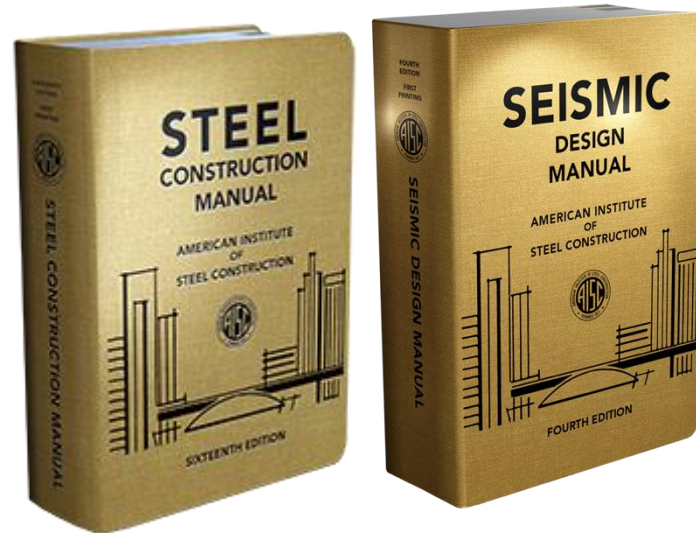
CE 123. Design of Reinforced Concrete Structures

Relevant Literature

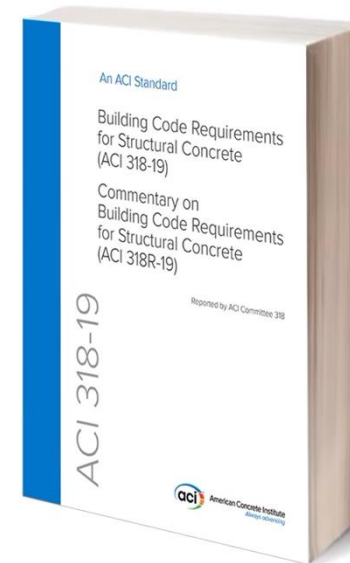
ASCE 7-22



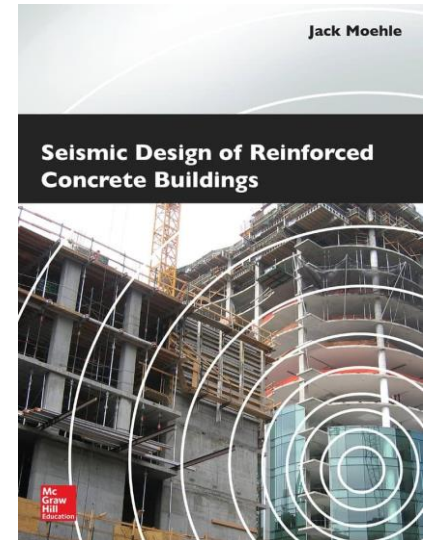
AISC Steel Manuals



ACI 318



Seismic Design of Reinforced Concrete Buildings



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